

Finding Side Lengths in Parallelograms

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A **parallelogram** is a quadrilateral with two pairs of parallel sides.

Opposite sides of a parallelogram are congruent
(equal in length).

Opposite Sides

If $ABCD$ is a parallelogram, then

$$AB = CD$$

and

$$BC = AD.$$

Finding Unknown Side Lengths

Algebraic expressions can be used to represent side lengths.

To find unknown side lengths, set opposite sides equal and solve the resulting equation.

Checking the Answer

After finding the value of the variable, substitute it back into the expression to determine the side measure.

Checking your answer helps verify that opposite sides have equal lengths.

Examples

Example 1

In parallelogram $ABCD$,

$$AB = 3x + 5, \quad CD = 5x - 9.$$

Since opposite sides are congruent,

$$3x + 5 = 5x - 9.$$

Solve for x and find AB .

Example 2

In parallelogram $WXYZ$,

$$WX = 6x - 4, \quad YZ = 3x + 17.$$

Since opposite sides are congruent,

$$6x - 4 = 3x + 17.$$

Solve for x and find WX .

Example 3

In parallelogram $RSTU$, the perimeter is 76 cm.

One side measures $2x + 3$ cm and the adjacent side measures $3x + 2$ cm.

Find x .

Practice Exercises

Practice 1

In parallelogram $ABCD$,

$$AB = 3x + 5, \quad CD = 5x - 9.$$

Find x and the length of AB .

Practice 2

In parallelogram $WXYZ$,

$$WX = 6x - 4, \quad YZ = 3x + 17.$$

Find x and the length of WX .

Practice 3

In parallelogram $EFGH$,

$$FG = 2x + 11, \quad EH = 5x - 10.$$

Find x and the length of EH .

Practice 4

In parallelogram $PQRS$,

$$PS = 4x + 3, \quad QR = 2x + 19.$$

Find x and the length of QR .

Practice 5

In parallelogram $RSTU$, the perimeter is 76 cm.

One side measures $2x + 3$ cm and the adjacent side measures $3x + 2$ cm.

Find x .

Activities

Activity 1

In parallelogram $PQRS$,

$$PQ = 5x - 6, \quad RS = 3x + 8.$$

Find x and the length of RS .

Activity 2

In parallelogram $JKLM$,

$$JK = \frac{x}{3} + 9, \quad LM = \frac{2x}{3} + 1.$$

Find x and the length of LM .

Activity 3

In parallelogram $ABCD$,

$$BC = 6x - 9, \quad AD = 4x + 7.$$

Find x and the length of BC .

Activity 4

In parallelogram $MNOP$, the perimeter is 60 cm.

The adjacent sides are

$$x + 5 \text{ cm} \quad \text{and} \quad 2x - 2 \text{ cm}.$$

Find x and the lengths of the sides.

Activity 5

In parallelogram $VWXY$,

$$VW = 2x + 4, \quad WX = x + 10.$$

If the perimeter is 60 cm, find x and the lengths of all sides.

Thank you!